GDB Cheat Sheet for CSAPP Bomb Lab

Directory: /data/users/huanghao/cs224n/hw/labs/02-bomblab/bomb
Style: AT&T assembly. Source operand first, destination operand second.

Start GDB

```
cd /data/users/huanghao/cs224n/hw/labs/02-bomblab/bomb
gdb ./bomb
set disassembly-flavor att
set debuginfod enabled off
break explode_bomb
run answers.txt
```

AT&T Syntax Rules

```
mov $0x402400,%esi      => %esi = 0x402400
sub $0x8,%rsp           => %rsp = %rsp - 8
add $0x4,%rbx           => %rbx = %rbx + 4
cmp %eax,0xc%rsp        => compare memory value with %eax
$0x10                   => immediate constant
%rax                    => register
0x8%rsp                 => memory at %rsp + 8
0x402470,%rax,8         => memory at 0x402470 + %rax * 8
```

x86-64 Function Arguments

```
1st argument: %rdi
2nd argument: %rsi
3rd argument: %rdx
4th argument: %rcx
5th argument: %r8
6th argument: %r9
Return value: %rax
Current instruction: %rip
Stack pointer: %rsp
```

Safety Rule

```
Always use: break explode_bomb
If GDB stops there, do not continue. Use bt, inspect, then quit.
```

Breakpoints, Running, and Disassembly

Breakpoints and Running

```
break phase_1           stop when phase_1 starts
b phase_1               short form
break *0x400f60         stop at exact address
break explode_bomb      safety breakpoint
info breakpoints        list breakpoints
delete 2                delete breakpoint number 2
disable 2               disable breakpoint number 2
enable 2                enable breakpoint number 2
run answers.txt         start with answers file
r answers.txt           short form
continue                continue to next breakpoint
c                       short form
quit                    exit GDB
q                       short form
```

Disassembly

```
disassemble phase_1     show assembly for phase_1
disas phase_1           short form
x/10i $rip              show 10 instructions at current location
x/20i 0x400efc          show 20 instructions from an address
layout asm              optional TUI assembly view
layout regs             optional TUI register view
Ctrl-x a                toggle TUI mode
```

Inspect Registers

```
info registers          print all registers
i r                     short form
p/x $rax                print %rax in hexadecimal
p/d $rax                print %rax in decimal
p/x $rsp+0x8            print address expression
```

Stepping

```
nexti or ni             execute one instruction, do not enter calls
stepi or si             execute one instruction, enter calls
finish                  run until current function returns
until *0x4010bd         run until exact address
```

Inspect Memory with x

General form: x/[count][format][unit] address

Common Examples

```
x/s 0x402400            show C string at address
x/s $rdi                show string pointed to by %rdi
x/6dw $rsp              show 6 four-byte signed decimals
x/6xw $rsp              show 6 four-byte hex words
x/8gx 0x402470          show 8 eight-byte hex values
x/20cb $rdi             show 20 one-byte characters
x/10i $rip              show 10 instructions
```

Formats

```
s    string
i    instruction
x    hexadecimal
d    signed decimal
u    unsigned decimal
c    character
```

Units

```
b    1 byte
h    2 bytes
w    4 bytes
g    8 bytes
```

Useful Bomb Lab Workflow

1. Put known answers in answers.txt, one phase per line.
2. Start GDB: gdb ./bomb
3. Set: set disassembly-flavor att
4. Set: break explode_bomb
5. Run: run answers.txt
6. Disassemble next phase: disas phase_3
7. Inspect strings, stack, jump tables with x/s, x/dw, x/gx.
8. If it stops at explode_bomb: bt, inspect, quit.

Bomb Lab Phase Patterns and Examples

Phase Patterns

```
phase_1: compare input with constant string; use x/s address
phase_2: read six numbers; use x/6dw $rsp
phase_3: read two numbers and use jump table; use x/8gx table
phase_4: recursive func4; track %edi, %esi, %edx
phase_5: six-character mapping; use x/s table and low 4 bits
phase_6: linked list reorder; inspect node memory
```

Examples from This Bomb

```
x/s 0x402400            phase_1 target string
x/s 0x4025cf            phase_3 and phase_4 format string
x/8gx 0x402470          phase_3 jump table
x/s 0x4024b0            phase_5 lookup table
x/s 0x40245e            phase_5 target string
```

When Something Goes Wrong

```
bt                      backtrace: see which phase exploded
x/s $rdi                at compare function, first string argument
x/s $rsi                at compare function, second string argument
cat -A answers.txt      in shell, check newlines and hidden chars
Ctrl-C                  interrupt a running program in GDB
```

High-value Commands to Memorize First

```
b explode_bomb
run answers.txt
disas phase_N
x/s ADDRESS
x/6dw $rsp
x/8gx ADDRESS
i r
ni, si, c, q
```